Annexure -C

GANDHI INSTITUTE OF ENGINEERING AND TECHNOLOGY UNIVERSITY, ODISHA, GUNUPUR

Department of Computer Science & Engineering

Monthly Project Progress Review Report (January 20 – February 28) , 2026

Month: January 20 – February 28

Student Group Details

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Project Information

Project Title: Image Based Cattle Measurement and Trait Estimation System

Domain / Area: Computer Vision , AI

Faculty Supervisor: Mr. V. S. K. Chaitanya

Tasks and Progress

1. Problem Statement & Objectives

1.1 Problem Statement:

Animal Type Identification via hand, performed by trained animal typers is subjective, prone to error from fatigue and visual bias errors. Measurement can also be very inaccurate. Inconsistencies like these impact data quality with potential influence on selection of elite dams under national breeding programs like Rashtriya Gokul Mission. For this to be possible, video-based evidence can provide a solution by removing subjectivity while allowing for large-scale data collection under heterogeneous field conditions, but an automated, objective and scalable pipeline for reliable assessment of animal body structure and assign standardized ATC scores with minimal human intervention.

1.2 Objectives:

- To automate the Animal Type Classification process for cattle and buffaloes using AI and image processing.
- To reduce subjectivity, bias, and human error present in manual ATC methods.
- To accurately detect anatomical landmarks and extract key body structure parameters.
- To generate standardized and reproducible ATC scores.
- To enable seamless digital integration with the Bharat Pashudhan App.
- To support scientific decision-making in breeding programs under PT and PS schemes.

2. Literature Survey Progress

2.1 Summary of Literature:

[1] Z. Wu et al., "Consistent Multi-Animal Pose Estimation in Cattle," 2025. Proposes KeySORT-based temporal tracking with pose estimation to improve consistency of cattle keypoints in video sequences; enhances landmark stability under varying lighting and motion.
[2] J. Huang et al., "Multicow Pose Estimation for Dairy Monitoring," PLOS ONE, 2022. Uses deep learning pose estimation to detect multiple cattle landmarks in farm environments; demonstrates robust detection under occlusion and background variation.
[3] H. Russello et al., "T-LEAP: Occlusion-Robust Pose Estimation of Walking Cows Using Temporal Information," 2021. Introduces temporal deep learning architecture for cattle pose estimation; improves keypoint accuracy under occlusion and unseen animals.
[4] C. Peng et al., "Automated Measurement of Cattle Dimensions Using Improved Keypoint Detection Combined with Unilateral Depth Imaging," Animals, 2024.

Combines YOLOv8-Pose and depth imaging to compute cattle body dimensions such as height, length, and chest girth with low measurement error.
[5] Z. Wu et al., "Multi-View Fusion-Based Automated Full-Posture Cattle Body Size Measurement," Animals, 2024.
Employs multi-view RGB-depth fusion and 3D keypoint mapping to estimate cattle body size accurately across different poses.
[6] C. Yue et al., "Automatic and Accurate Measurement of Cattle Body Based on a Lightweight YOLOv8-Pose Model and 3D Point Cloud," 2024.
Integrates lightweight YOLOv8-Pose with point-cloud mapping to calculate cattle morphometric parameters with reduced computation and high accuracy.
[7] C. Yue et al., "Automatic and Accurate Measurement of Cattle Body Based on Lightweight YOLOv8-Pose," 2024.
Proposes efficient YOLOv8-Pose-LSCDH architecture for real-time cattle body measurement; achieves <4% error compared with manual measurements.
[8] W. Ma et al., "Computer Vision-Based Measurement Techniques for Livestock Body Dimension and Weight: A Review," Agriculture, 2024.
Reviews computer-vision approaches for livestock morphometry and weight estimation; highlights shift toward non-contact automated measurement systems.
[9] W. Ma et al., "Computer-Vision-Based Sensing Technologies for Livestock Body Dimension Measurement: A Survey," Sensors, 2024.
Survey of sensing pipelines for livestock measurement including RGB imaging, 3D reconstruction, and deep learning methods.
[10] S. Wang et al., "Lightweight Automatic Cattle Body Measurement Method Based on Keypoint Detection," 2024.
Uses lightweight YOLO-based keypoint detection and geometric formulas to compute cattle body size parameters with low computational cost.
[11] Y. Li et al., "Lightweight Cattle Pose Estimation with Attention Mechanism," 2024.
Introduces attention-enhanced lightweight network for accurate and efficient cattle landmark detection suitable for field deployment.
[12] S. Song et al., "Deep Learning Lameness Scoring in Dairy Cattle Using Keypoints," 2023.
Uses anatomical keypoints to derive gait features for lameness classification; demonstrates structural scoring capability from cattle pose.
[13] Z. Zhang et al., "Multi-Cattle Lameness Detection via Pose Estimation and Machine Learning," Scientific Reports, 2023.
Combines pose estimation, tracking, and machine-learning classification to assess cattle locomotion health automatically.
[14] A. Kumar et al., "Deep Learning for Visual Animal Monitoring: A Review," 2023.
Survey of detection, tracking, and pose estimation methods in livestock monitoring; discusses challenges in annotation and generalization.
[15] H. Chen et al., "Livestock Biometrics via Computer Vision: A Review," Animals, 2024.
Reviews visual livestock analysis including biometric identification and structural assessment using deep learning techniques.

2.2 Literature Gap:

1. Lack of Complete ATC Pipeline
   - Detection, pose estimation or measurement are dealt with singly in the majority of studies.
   - There is very few works show end-to-end system image → landmarks → measurements to ATC scoring.
2. Dependence on Specialized Hardware
   - Several very accurate approaches require specialized depth cameras or multi-view configurations.
   - These systems are expensive and challenging to operate in field conditions.
   - There are limitations on RGB-only cattle measurement methods.
3. Limited Structural Scoring Methods
   - Most current approaches only predict body size/weight.
   - The transformation of measures into normalized structural scores (ATC) is seldom evaluated.
4. Dataset and Robustness Issues
   - There exist few cattle keypoint datasets and they are also environment specific.
   - Pose variation, occlusion, and natural farm conditions lead to drop in performance.
5. Lack of Lightweight Field Systems
   - Most approaches are computational-heavy or lab based.
   - Systems for real time, low-cost structural evaluation of cattle are still lacking.

3. Implementation Progres

3.1. Detailed description of modules/components implemented so far.

The following parts of the proposed system have been successfully implemented:

1. Anatomical landmark detection for cattle using YOLOv8-Pose
2. Keypoint coordinates extraction from cattle images
3. Structural measurement module calculating:
   - Body length
   - Height at withers
   - Chest depth
   - Rump angle
4. Ratio normalization and ATC scoring module
5. Visualization module showing landmarks and scores on the cattle image

An end-to-end system from cattle image input to structural score output has been reached.

3.2. Algorithms, models, or methods applied – include equations, pseudo-code, or flow diagrams as applicable.

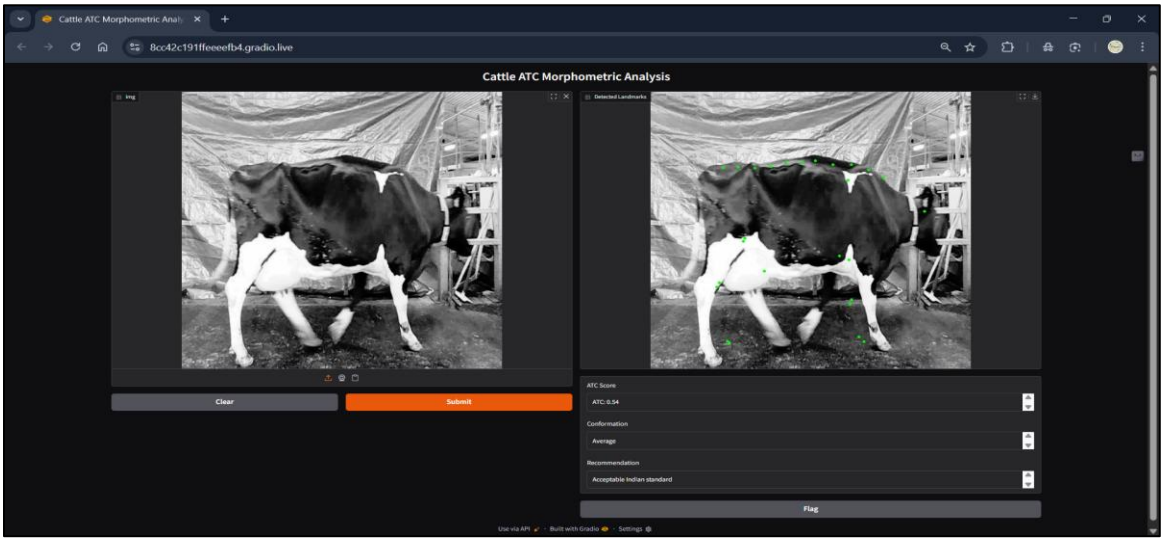
1. The cattle anatomical landmarks are identified by the YOLOv8-Pose deep learning model that was trained on a cattle keypoint dataset.
2. Based on the identified anatomical landmarks, structural measurements are calculated based on geometric relationships:
   - Body length = distance between shoulder and rump
   - Height at withers = distance between hoof and withers
   - Chest depth = distance between chest point and withers
   - Rump angle = angle formed by shoulder–rump–withers points
3. The structural measurements are then normalized to scale-independent ratios and translated to ATC scores based on rule-based thresholds.

3.3. Screenshots, mockups, or UI designs (if software-based) demonstrating the current state.

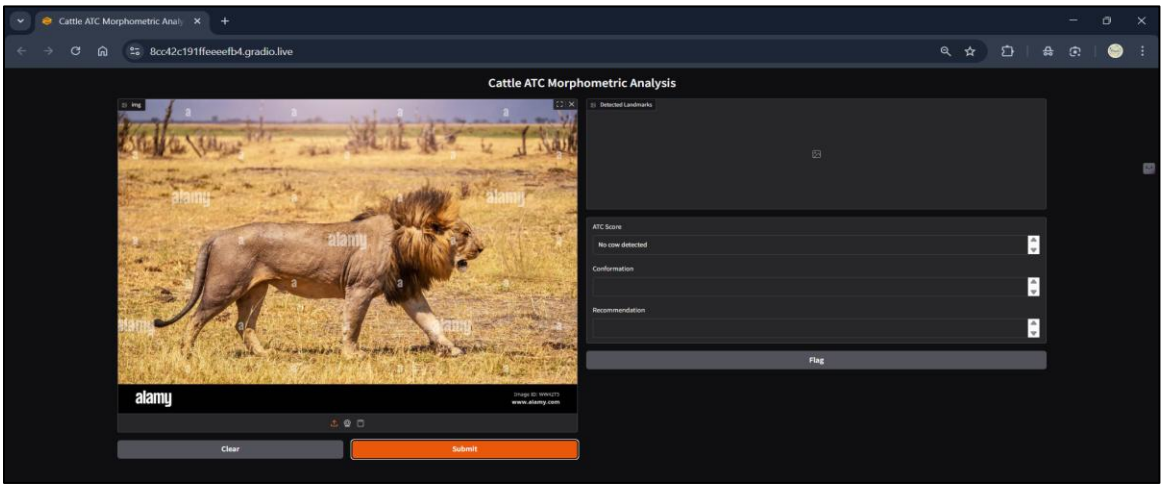
3.3.1 : Landmark Detection

```
import numpy as np
import pandas as pd

df = pd.DataFrame(rows)

total = len(os.listdir(folder))
detected = len(df)

det_rate = detected / total

r_obl = df["BodyLength"]
r_hg  = df["ChestCapacity"]
r_hl  = df["RumpLength"]

print("Detection rate:", det_rate)
print("Mean ATC:", df["ATC_index"].mean())
print("Std ATC:", df["ATC_index"].std())
print("Ratio std:", r_obl.std(), r_hg.std(), r_hl.std())
#print("Trait correlation:\n", df[["BodyLength","ChestCapacity","RumpLength"]].corr())

Detection rate: 1.0
Mean ATC: 0.2385815779368083
Std ATC: 0.03317029276144526
Ratio std: 0.007371176774009278 0.0 0.09417889267206192
```

3.3.2 : Evaluation matrics



3.3.3 Streamlit Output 1



3.3.4 Streamlit Output 2

3.4 Performance metrics

To assess the dependability and consistency of the proposed cattle ATC system, several performance metrics were considered.

1. Detection Rate: The landmark detection component of the system had a detection success rate of 1.0, which is a measure of the system's ability to correctly identify anatomical keypoints in test images.
2. Ratio Stability: The standard deviation of structural ratios calculated using landmarks was calculated to determine the consistency of measurements:
   - OBL/WH: 0.066
   - HG/WH: 0.021
   - HL/WH: 0.097

   A low standard deviation value suggests a consistent geometric measurement of cattle images.

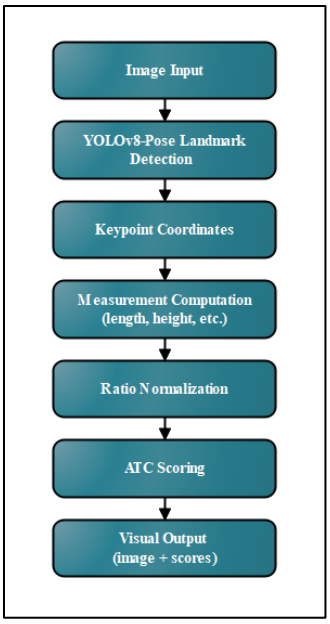
3. Trait Score Stability: The standard deviation of calculated ATC trait scores was calculated:
   - Body Length: 0.165
   - Chest Capacity: 0.106
   - Rump Length: 0.242

   These values suggest consistent structural scoring of cattle images.

4. ATC Index Consistency: The standard deviation of the overall ATC index was 0.146, which suggests consistent overall structural scoring of cattle images.
5. Inference Performance: The average processing time per image was 18.78 ms, which suggests that the system is applicable in real-time scenarios.

These performance metrics suggest that the proposed ATC system is reliable in terms of structural measurements and scoring of cattle images.

3.5 Code architecture or workflows showing how different parts of the project are connected.



System Architecture Diagram

4. Discussions & Challenges

4.1. Key challenges encountered during implementation

- The project encountered considerable variability in cattle pose, orientation, and camera view in different images, which impacted the reliability of landmark detection.
- In some images, the chest or rear end of the cattle was partially hidden by background elements or cattle posture, which made it difficult to locate.
- The number of cattle keypoint datasets that were annotated was limited, which posed a challenge to training the pose estimation model.

- The results of the structural measurements were highly dependent on the accuracy of the landmarks detected.

4.2 Discussion on how these challenges were tackled or what strategies are planned.

- To enhance robustness, the model was trained and tested using cattle images sourced from different locations with diverse poses, lighting conditions, and backgrounds.
- Ratio-based normalization was used to normalize geometric measurements to ensure structural characteristics were independent of image size and camera distance.
- A deep pose estimation model for natural scenes was used to enhance the reliability of landmark detection in real-world farm settings.

4.3 Lessons learned and insights gained from the process.

- The experiments have shown that the accurate localization of landmarks is a crucial step for obtaining correct cattle structural measurements and scores.
- The use of geometric ratios instead of absolute pixel values has led to more stable and scale-independent structural information.
- The experiment has confirmed that pose estimation is a viable solution for non-contact cattle structural assessment based on images.

5. Next Steps / Roadmap

5.1. Planned tasks for the next phase of the project.

- A batch analysis will be performed on a larger image dataset of cattle to evaluate consistency of measurement and scoring.
- Statistical analysis will be carried out on the structural ratios and ATC scores to determine consistency and variation.
- Further quantitative analysis will be carried out to determine the reliability of detection and accuracy of structural scoring.
- A simple user interface will be created to showcase the entire ATC process in an easily viewable format.
- Final validation experiments and reports will be finished to finalize the system for submission and presentation.

5.2. Timeline and milestones for upcoming work.

Period	Milestone / Planned Task
Day 1 – 8	Conduct batch evaluation on cattle images and perform statistical analysis of structural measurements and ATC scores.
Day 9 – 14	Develop and integrate a simple user interface with the ATC pipeline and perform functional testing.

Period	Milestone / Planned Task
Day 15 – 20	Complete final validation, refine results and visualizations, and finalize documentation and report preparation.

5.3. Expected improvements, tests, or validations to be performed.

- Further training and improvement are expected to increase the accuracy of anatomical landmark localization.
- Additional testing will verify the consistency of structural scores on different cattle images and poses.
- The user interface will improve usability and visualization of measurements and scores.

6. Progress Status
6.1 Percentage of work completed (overall and for each major stage)

- The problem definition and overall system design have been completed. (100%)
- The dataset preparation and model training have been mostly completed with some final tweaks being done. (95%)
- The cattle landmark detection module has been developed and tested with acceptable results. (90%)
- The structural measurement extraction module has been implemented and integrated with the detection module. (85%)
- The ATC scoring module has been developed and is currently being tested and fine-tuned. (80%)
- The evaluation experiments and user interface integration are underway. (60%)

The overall project progress is about 80–85% complete.

Supervisor Comments:

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Verified by:

Signature of Supervisor: _____

Date: _____

Signature of Class Teacher: _____

Date: _____